

**HOW TO USE THIS SET**

20 questions, mixed difficulty, 3 case-based. Attempt every question before checking the answer key on the last page.

4 Easy

10 Medium

6 Hard

1.**EASY**

Which structure is found at each of the six corners of a classic hepatic lobule?

- A. Central vein
- B. Portal triad
- C. Islet of Langerhans
- D. Kupffer cell cluster
- E. Ito cell nest

2.**EASY**

Which liver cell resides in the space of Disse, stores vitamin A, and becomes a major collagen source during liver injury?

- A. Kupffer cell
- B. Ito (hepatic stellate) cell
- C. Hepatocyte
- D. Sinusoidal endothelial cell
- E. Cholangiocyte

3.**EASY**

Kupffer cells are best described as:

- A. Monocyte-derived resident macrophages on the luminal surface of the sinusoid endothelium, phagocytosing aged RBCs and debris
- B. The main structural cell of the hepatic lobule
- C. The primary bile-producing cell
- D. Cells that store vitamin A
- E. Hormone-secreting endocrine cells of the liver

4.**EASY**

The wall of the bile canaliculus is formed by:

- A. The plasma membranes of two adjacent hepatocytes, with no distinct duct wall
- B. Kupffer cell processes
- C. A distinct simple cuboidal epithelium
- D. Fenestrated endothelium
- E. Connective tissue septa

5.**MEDIUM**

Which vessel supplies the majority (~80%) of the liver's blood flow, and what is notable about this blood?

- A. Inferior vena cava; mixed venous blood
- B. Hepatic artery; highly oxygenated
- C. Portal vein; oxygen-poor but nutrient-rich, from the pancreas, spleen and intestine
- D. Hepatic vein; deoxygenated systemic blood
- E. Splenic artery; oxygen-rich

6.

MEDIUM

In the hepatic acinus model of liver zonation, Zone I (periportal) compared with Zone III (pericentral) is:

- A. Devoid of hepatocytes
- B. Lower in oxygen and nutrients
- C. The most ischaemia-prone zone
- D. Identical in oxygen content to Zone III
- E. Higher in oxygen, nutrient and toxin concentration, since it is nearest the portal triad

7.

MEDIUM

Which zone of the hepatic acinus is most vulnerable to ischaemia, being furthest from the oxygen-rich portal blood supply?

- A. Zone II exclusively
- B. All zones are equally vulnerable
- C. Zone I
- D. Zone III (nearest the central vein)
- E. The portal triad itself

8.

MEDIUM

Bile release from the gallbladder is triggered mainly by which hormone?

- A. Gastrin
- B. Glucagon
- C. Somatostatin
- D. Secretin
- E. Cholecystokinin (CCK)

9.

MEDIUM

The three overlapping models used to subdivide liver microarchitecture — classic lobule, portal lobule, and hepatic acinus — are each based on:

- A. Direction of blood flow, direction of bile flow, and the oxygen/nutrient/toxin gradient respectively
- B. Kupffer cell density
- C. Direction of lymphatic flow only
- D. Distance from the gallbladder
- E. Hepatocyte ploidy

10.

MEDIUM

The classic hepatic lobule is:

- A. Circular, with no defined landmarks
- B. Hexagonal, with a central vein at the centre and portal triads at the six corners
- C. Irregular, defined only by hepatocyte plates
- D. Triangular, with a portal triad at the centre
- E. Diamond-shaped, with two central veins

11.

MEDIUM

Which hepatocyte surface forms the wall of the bile canaliculus and is the primary site of bile secretion?

- A. Sinusoidal surface
- B. Basal surface facing the capsule
- C. There is no specialised surface for bile secretion
- D. Canalicular surface
- E. Intercellular surface

12.

MEDIUM

Correctly order the bile duct pathway from the bile canaliculus toward the duodenum.

- A. Bile canaliculus → sinusoid → space of Disse → duodenum
- B. Bile canaliculus → hepatic artery → portal vein → duodenum
- C. Bile canaliculus → central vein → hepatic vein → duodenum
- D. Bile canaliculus → common bile duct → canal of Hering → bile duct → duodenum
- E. Bile canaliculus → canal of Hering (bile ductule) → bile duct → hepatic duct → common bile duct → ampulla of Vater → duodenum

13.

MEDIUM

Approximately what proportion of bile acids are reabsorbed from the intestine rather than newly synthesised by hepatocytes?

- A. 0%, all are synthesised fresh
- B. 50%
- C. 100%, none are reabsorbed
- D. 90%
- E. 10%

14.

MEDIUM

The gallbladder mucosa is characterised by:

- A. Simple squamous epithelium only
- B. Goblet cells interspersed with absorptive cells
- C. A thick keratinized surface
- D. Highly folded simple columnar epithelium with microvilli and no goblet cells
- E. Stratified squamous epithelium

15.

HARD CASE

A patient with cirrhosis has extensive fibrosis on liver biopsy, with excess collagen deposition around sinusoids identified on trichrome stain.

Which cell type is most directly responsible for this excess collagen, based on its described role in liver injury?

- A. Ito (hepatic stellate) cell, which becomes the major source of collagen/ECM during liver injury
- B. Sinusoidal endothelial cell
- C. Kupffer cell
- D. Centroacinar cell
- E. Hepatocyte

16.

HARD

In acetaminophen (paracetamol) toxicity, hepatocyte necrosis characteristically begins in the pericentral (Zone III) region. Which explanation, based on acinar zonation, best accounts for this pattern?

- A. Zone III is furthest from the oxygen- and nutrient-rich portal blood supply, making it the most vulnerable to toxic and ischaemic injury, consistent with it being the last zone perfused
- B. Zone I is always more vulnerable to any hepatotoxin
- C. Zonation has no bearing on injury pattern
- D. Zone III hepatocytes have no exposure to portal blood at all
- E. Zone III hepatocytes lack mitochondria and cannot metabolise the toxin

17.

HARD CASE

A gallstone lodges in the common bile duct, causing complete obstruction. Bile that would normally reach the duodenum backs up.

Tracing the pathway in reverse from the point of obstruction, which structure would show the earliest and most direct upstream backup?

- A. The intrahepatic bile ducts and canals of Hering, since bile flows canaliculus → canal of Hering → bile duct → hepatic duct → common bile duct, and obstruction at the common bile duct backs pressure up this entire chain
- B. The portal vein directly
- C. The hepatic artery
- D. The central vein of the hepatic lobule
- E. The pancreatic islets

18.

HARD

A researcher wants to visualise the portal lobule model rather than the classic lobule on a liver section. What landmark arrangement should they look for?

- A. The same hexagonal arrangement as the classic lobule, viewed from a different angle
- B. No landmarks are needed, since the portal lobule is not a real anatomical model
- C. A hexagonal outline with the central vein at the centre
- D. Interlocking triangles with a portal triad at the centre and a central vein at each of the three corners — the inverse landmark arrangement of the classic lobule
- E. A diamond shape with two central veins and two portal triads

19.

HARD CASE

A liver biopsy shows sinusoids lined by discontinuous, fenestrated endothelium, with a clear space between the endothelium and the hepatocyte surface, into which hepatocyte microvilli project.

What is this space called, and what is its functional significance?

- A. The portal canal, containing the portal triad
- B. The subcapsular space beneath Glisson's capsule
- C. The perisinusoidal duct, part of the biliary tree
- D. The space of Disse, which allows efficient exchange of materials between sinusoidal blood and hepatocytes via the hepatocyte microvilli
- E. The bile canaliculus, where bile is secreted

20.

HARD

Which statement correctly integrates the roles of Kupffer cells and Ito cells in the sinusoidal microenvironment during chronic liver injury?

- A. Kupffer cells are responsible for all hepatic collagen production
- B. Kupffer cells store vitamin A while Ito cells phagocytose debris
- C. Ito cells are macrophage-derived, like Kupffer cells
- D. Neither cell type interacts with the other
- E. Kupffer cells clear debris/pathogens and can activate Ito cells via cytokines, while Ito cells respond by increasing collagen production and can regulate sinusoidal blood flow through their contractile activity

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	B	Portal triad
2	B	Ito (hepatic stellate) cell
3	A	Monocyte-derived resident macrophages on the luminal surface of the sinusoid endothelium, phagocytosing aged RBCs and debris
4	A	The plasma membranes of two adjacent hepatocytes, with no distinct duct wall
5	C	Portal vein; oxygen-poor but nutrient-rich, from the pancreas, spleen and intestine
6	E	Higher in oxygen, nutrient and toxin concentration, since it is nearest the portal triad
7	D	Zone III (nearest the central vein)
8	E	Cholecystokinin (CCK)
9	A	Direction of blood flow, direction of bile flow, and the oxygen/nutrient/toxin gradient respectively
10	B	Hexagonal, with a central vein at the centre and portal triads at the six corners
11	D	Canalicular surface
12	E	Bile canaliculus → canal of Hering (bile ductule) → bile duct → hepatic duct → common bile duct → ampulla of Vater → duodenum
13	D	90%
14	D	Highly folded simple columnar epithelium with microvilli and no goblet cells
15	A	Ito (hepatic stellate) cell, which becomes the major source of collagen/ECM during liver injury
16	A	Zone III is furthest from the oxygen- and nutrient-rich portal blood supply, making it the most vulnerable to toxic and ischaemic injury, consistent with it being the last zone perfused
17	A	The intrahepatic bile ducts and canals of Hering, since bile flows canaliculus → canal of Hering → bile duct → hepatic duct → common bile duct, and obstruction at the common bile duct backs pressure up this entire chain
18	D	Interlocking triangles with a portal triad at the centre and a central vein at each of the three corners — the inverse landmark arrangement of the classic lobule
19	D	The space of Disse, which allows efficient exchange of materials between sinusoidal blood and hepatocytes via the hepatocyte microvilli
20	E	Kupffer cells clear debris/pathogens and can activate Ito cells via cytokines, while Ito cells respond by increasing collagen production and can regulate sinusoidal blood flow through their contractile activity

Histology GIT Block · MEDucation by Mattin Majid · Source: GIT Histology SGL 4, Dr Hero K. Mustafa

Recalled and source questions are treated as evidence of what gets asked, not as a source of correct answers — verify against your lecture notes.